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E^{2}AD: Enhanced and explainable Alzheimer's disease detection framework via anatomy- and relation-aware cross-modal knowledge distillation

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative disorder for which MRI and PET provide complementary structural and molecular information. Yet PET remains costly and often unavailable, motivating MRI-only diagnostic systems that still benefit from multimodal supervision. Existing methods either synthesize PET from MRI or limit cross-modal learning to low-dimensional spaces, underutilizing MRI–PET complementarity and leading to limited robustness and generalizability. To address these challenges, we introduce E^{2}AD, an **E**nhanced and **E**xplainable **AD** detection framework that leverages anatomy- and relation-aware cross-modal knowledge distillation (KD). Using paired MRI–PET data during training but only MRI at inference, E^{2}AD augments traditional logit-based KD through two synergistic components: (1) *anatomy-aware distillation* that transfers within-subject anatomical dependencies through an anatomical Mixture-of-Mappers, yielding spatially meaningful and clinically traceable cues; and (2) *relation-aware distillation* that promotes stable between-subject structural relations through generalizable pairwise alignment, yielding a representation space with better cross-cohort generalization. To enhance clinical utility, we further introduce a tailored multi-agent workflow that translates E^{2}AD's anatomical attention into structured, clinician-oriented MRI reports. Extensive experimental results on the internal ADNI cohort and two external cohorts (AIBL and NACC) demonstrate that E^{2}AD outperforms state-of-the-art baselines, offering faster convergence, improved data efficiency, stronger cross-cohort generalization, and enhanced explainability. Source code is available at https://github.com/thibault-wch/E2AD-for-Alzheimer-disease.

1. Introduction

Alzheimer's disease (AD) affects over 55 million people and is projected to surpass 150 million by 2050 (Alzheimer's Association, 2024). Early detection and longitudinal monitoring are pivotal for timely intervention, yet AD's progressive and heterogeneous neuropathology poses challenges to reliable assessment in routine clinical practice.

Among neuroimaging modalities, magnetic resonance imaging (MRI) is widely accessible and captures structural atrophy, whereas

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[1] Data used in preparation of this article was obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (https://adni.loni.usc.edu). The investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

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positron emission tomography (PET) provides pathophysiological readouts. In principle, both imaging modalities are complementary; however, PET is often unavailable due to access constraints, radiation exposure, and cost. Current deep-learning pipelines gravitate toward two extremes: (1) *multimodality models* (Yang et al., 2022; Qiu et al., 2024; H. Zhou et al., 2024; Zhang et al., 2024) that presuppose full-modality inference-complete on curated datasets but are brittle when PET is absent, as often occurs in clinical workflows; and (2) *single-modality models* (Liu et al., 2018; Jabason et al., 2024; C. Wang et al., 2024; Song et al., 2026) that embrace the MRI-only input and thus avoid missing data, but leave complementary cross-modal signal untapped, limiting differential diagnosis.

Recent efforts to bridge this discrepancy fall into two lines. First, MRI-to-PET scan synthesis for modality completion (Pan et al., 2020; Ou et al., 2024; Wang et al., 2024; Xu et al., 2025) prioritizes image fidelity and therefore only indirectly supports clinical discrimination. The second learns cross-modal differences in a low-dimensional space, *e.g.* logits or global features (Guan et al., 2021; Chen et al., 2023; Yang et al., 2024; Kwak et al., 2025; Liu et al., 2025), yet fails to capture disease-relevant anatomy and cohort-level knowledge. As a result, both strategies underuse multimodal supervision, yielding overconfident predictions and brittle generalization.

To address this, we propose E^{2}AD, an **E**nhanced and **E**xplainable **AD** detection framework via anatomy- and relation-aware knowledge distillation (KD). Trained with paired MRI+PET yet requiring only MRI at deployment (Fig. 1), E^{2}AD operationalizes a simple principle: learning *where* diagnostic evidence resides within a case and *how* cases should be organized across the cohort. Specifically, E^{2}AD extends traditional logit-based KD (*LogitKD*) via two synergistic components at orthogonal levels of diagnostic knowledge: (1) anatomy-aware KD (*AnaKD*), which transfers *within-subject* anatomical weights from teacher to student via an anatomical Mixture-of-Mappers (A-MoM) with ROI-specific and ROI-shared mappers, yielding spatially meaningful, clinically traceable cues; and (2) relation-aware KD (*RelKD*), which fosters stable *between-subject* structural consistency by aligning student-related relations to the teacher–teacher relation, yielding better cross-cohort generalization. Additionally, we propose a tailored 3D MRI report agent workflow that provides detailed analysis towards practical utility.

The contributions of this work are summarized as follows.

1. We propose E^{2}AD, a cross-modal KD framework that aligns anatomy- and relation-aware capabilities from an MRI+PET multimodal teacher to an MRI-only student, enabling enhanced and explainable AD assessment.
2. We propose anatomy-aware distillation, which distills within-subject anatomical weights via an A-MoM, yielding spatially meaningful and clinically traceable cues.
3. We propose relation-aware distillation, which promotes stable between-subject structural relations through generalizable pairwise alignment, yielding a representation space with better cross-cohort generalization.
4. We propose a tailored MRI report multi-agent workflow that converts E^{2}AD's anatomical weights and demographic context into clinician-facing, region-level narratives for decision support.
5. Extensive experiments on ADNI and two external cohorts (AIBL and NACC) show that E^{2}AD achieves state-of-the-art (SOTA) performance with faster convergence, improved data efficiency, and enhanced explainability.

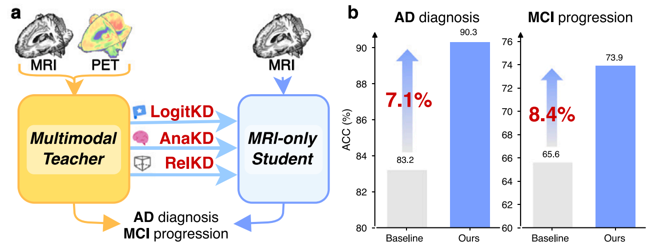

Fig. 1. Conceptual illustration and performance of E^{2}AD. (a) E^{2}AD augments logit-level KD (*LogitKD*) with anatomy-aware (*AnaKD*) and relation-aware (*RelKD*), trained on MRI–PET pairs but using only MRI at inference. **(b)** Performance improvement of E^{2}AD over the baseline.

2. Methodology

To enhance and explain AD detection, we propose an anatomy- and relation-aware cross-modal KD framework that transfers multimodal knowledge to MRI (Fig. 2**(a)**). Section 2.1 formalizes conventional KD in our setting; Section 2.2 and Section 2.3 describe the proposed anatomy- and relation-aware distillations, respectively; Section 2.4 summarizes E^{2}AD; and Section 2.5 outlines a tailored multi-agent workflow that converts E^{2}AD's anatomical routing weights into clinician-facing MRI reports.

2.1. Problem formulation and conventional KD

We define two training partitions: (1) a multimodal dataset $\mathcal{D}_{\text{Multi}} = \{(\boldsymbol{X}_i^{\text{M}}, \boldsymbol{X}_i^{\text{P}}, y_i)\}_{i=1}^{N_1}$, where each subject has an MRI $\boldsymbol{X}_i^{\text{M}} \in \mathbb{R}^{1\times D\times H\times W}$, a PET scan $\boldsymbol{X}_i^{\text{P}} \in \mathbb{R}^{1\times D\times H\times W}$, and a diagnostic label y_i; and (2) an MRI-only dataset $\mathcal{D}_{\text{Single}} = \{(\boldsymbol{X}_j^{\text{M}}, y_j)\}_{j=1}^{N_2}$. Let $\mathcal{T}$ and $\mathcal{S}$ be the multimodal teacher and MRI-only student, yielding logits $\hat{\boldsymbol{p}}^{\mathcal{T}}, \hat{\boldsymbol{p}}^{\mathcal{S}} \in \mathbb{R}^k$ for k classes.

Standard KD combines a hard-label cross-entropy (CE) loss on all $N_1 + N_2$ MRI-available subjects with a logit-level alignment term on the N_1 paired subjects:

$$\mathcal{L}_{\text{logitKD}} = -\text{SM}(\hat{\boldsymbol{p}}^{\mathcal{T}})\log(\text{SM}(\hat{\boldsymbol{p}}^{\mathcal{S}})), \tag{1}$$

$$\mathcal{L}_{\text{KD}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{logit}}\mathcal{L}_{\text{logitKD}}, \tag{2}$$

where SM denotes the softmax and λ_{logit} weights the loss terms. Meanwhile, feature-level KD represents another mainstream paradigm that aligns pre-logit features between teacher and student via L1/L2 losses.

Nevertheless, both logit- and feature-level KD compress multimodal supervision into low-dimensional representations, underutilizing rich multimodal signals; in contrast, E^{2}AD adds complementary anatomy- and relation-aware distillations.

2.2. Anatomy-aware distillation

AD is characterized by widespread cerebral atrophy, with regionally heterogeneous progression across brain ROIs (Holland et al., 2009; Poulakis et al., 2018). To leverage this clinical prior, we align teacher–student *anatomical* knowledge across ROIs. We operationalize this idea with anatomical Mixture-of-Mappers (A-MoM), organized into three steps: (1) ROI tokenization, (2) anatomical mapper extraction, and (3) anatomy-aware fusion and distillation, as shown in Fig. 2**(b)**.

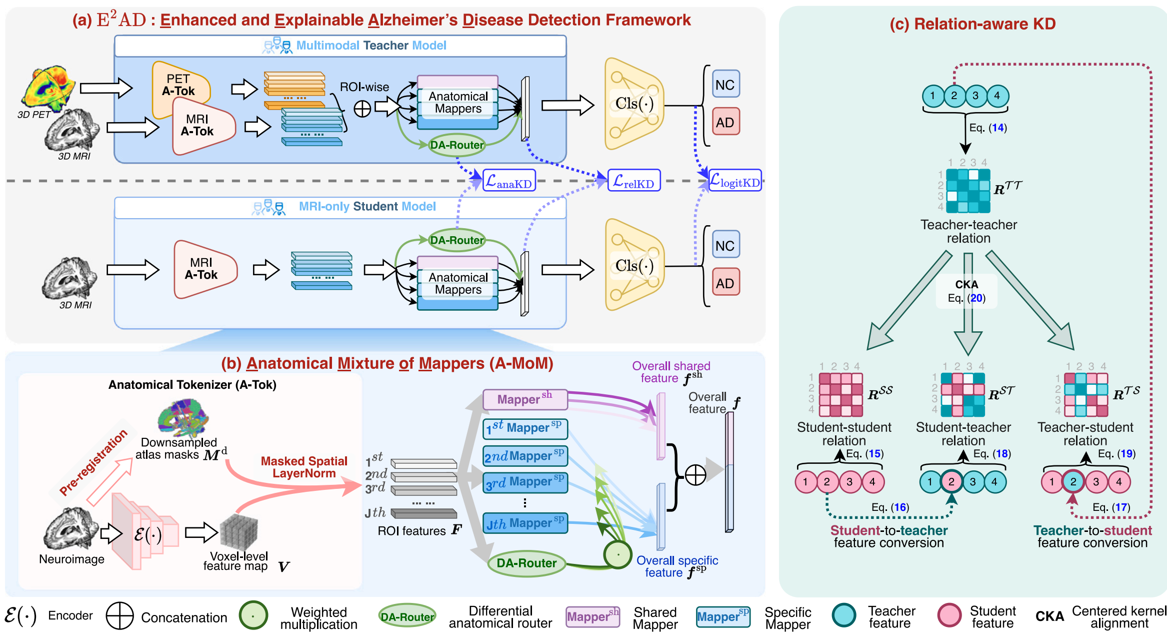

Fig. 2. **Detailed illustration of the proposed E²AD.** (**a**) Overall framework architecture. (**b**) Anatomy-aware distillation employs an anatomical mixture-of-mappers to tokenize ROIs, route features to specialized mappers, and achieve anatomy-aware fusion and distillation. (**c**) Relation-aware distillation aligns diverse student-related relations with the teacher–teacher relation, producing a more stable structural organization.

2.2.1. ROI tokenization

To replace conventional undifferentiated voxel-wise features with anatomically informed representations, we introduce an anatomical tokenizer (A-Tok) that converts a 3D feature map $\boldsymbol{V} \in \mathbb{R}^{C\times d\times h\times w}$ produced by an encoder $\mathcal{E}(\cdot)$ to J ROI features, where C, d, h, and w are channels, depth, height, and width, respectively. Given a downsampled, pre-registered atlas mask $\boldsymbol{M}^{\mathrm{d}}$ indexing ROI voxels, we flatten $\boldsymbol{V}$ over space to $\boldsymbol{V}^{\mathrm{flat}} \in \mathbb{R}^{C\times(dhw)}$ and form a binary ROI-voxel assignment matrix $\boldsymbol{M}^{\mathrm{d}} \in \{0,1\}^{J\times(dhw)}$ with $(\boldsymbol{M}^{\mathrm{d}})_{j,v} = 1$ iff voxel v belongs to ROI j. Let $D = \mathrm{diag}(\boldsymbol{M}^{\mathrm{d}}\mathbf{1})$ be the diagonal matrix of per-ROI voxel counts ($\mathbf{1}$ is an all-ones vector of length dhw). ROI features are masked averages with row-wise layer normalization LN(·):

$$\boldsymbol{F} = \mathrm{LN}\left(\frac{\boldsymbol{M}^{\mathrm{d}}(\boldsymbol{V}^{\mathrm{flat}})^{\top}}{D}\right), \tag{3}$$

where $\top$ denotes the matrix transpose.

2.2.2. Anatomical mapper extraction

Instantiation. We instantiate A-MoM with J ROI-specific mappers $\{\mathrm{Mapper}_j^{\mathrm{sp}}(\cdot)\}_{j=1}^{J}$ and an ROI-shared mapper $\mathrm{Mapper}^{\mathrm{sh}}(\cdot)$, all implemented as multilayer perceptrons (MLPs). Each ROI-specific mapper maps its ROI features to a C-dimensional vector; stacking yields ROI-specific features $\boldsymbol{F}^{\mathrm{sp}} \in \mathbb{R}^{J\times C}$. In parallel, the ROI-shared mapper jointly processes all ROI features to produce ROI-shared features $\boldsymbol{F}^{\mathrm{sh}} \in \mathbb{R}^{J\times C}$, capturing *global, ROI-shared disease signals* (*e.g.* overall atrophy burden or diffuse patterns) and amplifying discriminability among ROI-specific features. Formally,

$$\boldsymbol{F}^{\mathrm{sp}} = \mathtt{conc}(\mathrm{Mapper}_1^{\mathrm{sp}}(\boldsymbol{F}_{1,:}), \dots, \mathrm{Mapper}_J^{\mathrm{sp}}(\boldsymbol{F}_{J,:})), \tag{4}$$

$$\boldsymbol{F}^{\mathrm{sh}} = \mathrm{Mapper}^{\mathrm{sh}}(\boldsymbol{F}), \tag{5}$$

where conc denotes the column-wise (vertical) concatenation.

Mapper Regulation. To better regulate mappers, we propose mapper regulation loss $\mathcal{L}_{\mathrm{Reg}}$ that encourages similarity among ROI-shared features, disentangles ROI-specific ones, and ensures reconstruction fidelity, as shown in Fig. 3.

Define the overall shared feature as the ROI mean $\boldsymbol{f}^{\mathrm{sh}} = \frac{1}{J}\mathbf{1}^{\top}\boldsymbol{F}^{\mathrm{sh}}$, where $\mathbf{1} \in \mathbb{R}^{J}$ is the all-ones vector. The shared loss $\mathcal{L}_{\mathrm{sh}}$ aligns each

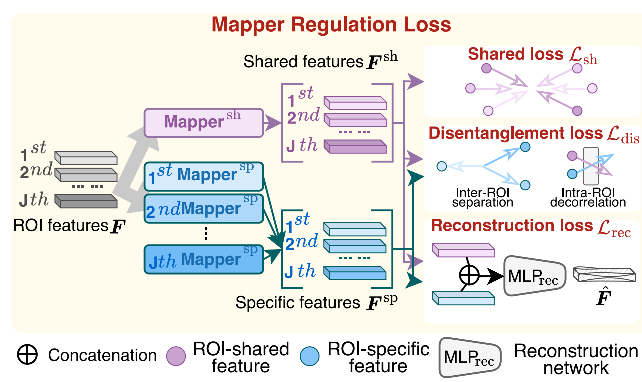

Fig. 3. Illustration of the mapper regulation loss.

ROI-shared feature to $\boldsymbol{f}^{\mathrm{sh}}$:

$$\mathcal{L}_{\mathrm{sh}} = \frac{1}{J}\sum_{j=1}^{J}\left(1 - \mathrm{Cos}(\boldsymbol{F}_{j,:}^{\mathrm{sh}}, \boldsymbol{f}^{\mathrm{sh}})\right), \tag{6}$$

where $\mathrm{Cos}(\cdot,\cdot)$ is the cosine similarity.

The disentanglement loss $\mathcal{L}_{\mathrm{dis}}$ encourages both inter-ROI separation of ROI-specific features and intra-ROI decorrelation between shared and specific features:

$$\mathcal{L}_{\mathrm{dis}} = \frac{1}{J^2}\Big(\underbrace{\textstyle\sum_{j_1\neq j_2}^{J}|\mathrm{Cos}(\boldsymbol{F}_{j_1,:}^{\mathrm{sp}}, \boldsymbol{F}_{j_2,:}^{\mathrm{sp}})|}_{\text{inter-ROI separation}} + \underbrace{\textstyle\sum_{j}^{J}|\mathrm{Cos}(\boldsymbol{F}_{j,:}^{\mathrm{sh}}, \boldsymbol{F}_{j,:}^{\mathrm{sp}})|}_{\text{intra-ROI decorrelation}}\Big), \tag{7}$$

where $|\cdot|$ denotes the absolute value function.

The reconstruction loss $\mathcal{L}_{\mathrm{rec}}$ ensures information fidelity by minimizing the reconstruction error with a network $\mathrm{MLP}_{\mathrm{rec}}(\cdot)$:

$$\mathcal{L}_{\mathrm{rec}} = \left\|\boldsymbol{F} - \mathrm{MLP}_{\mathrm{rec}}\left([\boldsymbol{F}^{\mathrm{sh}}, \boldsymbol{F}^{\mathrm{sp}}]\right)\right\|^2. \tag{8}$$

Thus, the mapper regulation loss $\mathcal{L}_{\mathrm{Reg}}$ is defined as:

$$\mathcal{L}_{\mathrm{Reg}} = \mathcal{L}_{\mathrm{sh}} + \mathcal{L}_{\mathrm{dis}} + \mathcal{L}_{\mathrm{rec}}. \tag{9}$$

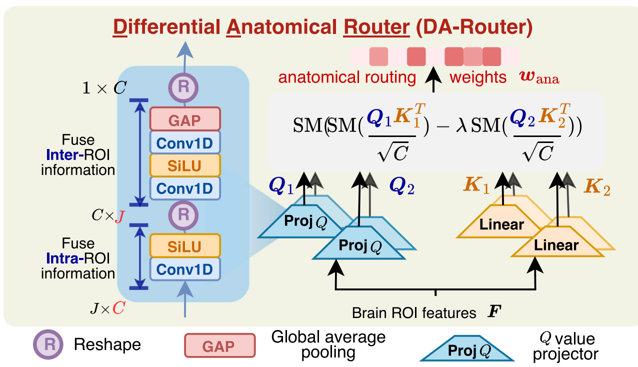

Fig. 4. Illustration of the differential anatomical router module.

2.2.3. Anatomy-aware fusion and distillation

To learn inter-ROI dependencies and enhance the explainability, we propose a differential anatomical router (DA-Router) module that assigns anatomical routing weights to adaptively route and fuse ROI-specific features, as shown in Fig. 4. Specifically, we first project the ROI features $\boldsymbol{F}$ into global queries $\boldsymbol{Q}_1, \boldsymbol{Q}_2 \in \mathbb{R}^C$ and ROI keys $\boldsymbol{K}_1, \boldsymbol{K}_2 \in \mathbb{R}^{J\times C}$. The DA-Router then computes anatomical weights as a probability distribution $\boldsymbol{w}_{\text{ana}} = \mathrm{SM}(\cdot) \in \Delta^{J-1}$ via a differential architecture. By computing the difference between attention maps rather than relying on absolute scores, this approach effectively suppresses vanilla attention noise (Vaswani et al., 2017) and facilitates a sharper focus on discriminative information (Ye et al., 2025; Cao et al., 2026):

$$\begin{aligned}\boldsymbol{w}_{\text{ana}} &= \text{DA-Router}(\boldsymbol{F}) \\ &= \mathrm{SM}\left(\mathrm{SM}(\frac{\boldsymbol{Q}_1\boldsymbol{K}_1^\top}{\sqrt{C}}) - \lambda \mathrm{SM}(\frac{\boldsymbol{Q}_2\boldsymbol{K}_2^\top}{\sqrt{C}})\right), \\ &\text{where} \quad [\boldsymbol{Q}_1;\boldsymbol{Q}_2] = \mathrm{Proj}_Q(\boldsymbol{F}), \quad [\boldsymbol{K}_1;\boldsymbol{K}_2] = \boldsymbol{F}\boldsymbol{W}^K.\end{aligned} \tag{10}$$

Here, $\boldsymbol{W}^K \in \mathbb{R}^{C\times 2C}$ is ROI key parameters, λ is a learnable scalar, and $\mathrm{Proj}_Q(\cdot)$ is the global query projector that integrates intra- and inter-ROI information to derive global scan information. To synchronize the learning dynamics of different query and key projectors, we re-parameterize the scalar λ as:

$$\lambda = \exp(\lambda_{\mathrm{q}_1}^\top \lambda_{\mathrm{k}_1}) - \exp(\lambda_{\mathrm{q}_2}^\top \lambda_{\mathrm{k}_2}) + \lambda_{\text{init}}, \tag{11}$$

where $\lambda_{\mathrm{q}_1}, \lambda_{\mathrm{q}_2}, \lambda_{\mathrm{k}_1}, \lambda_{\mathrm{k}_2} \in \mathbb{R}^C$ are learnable vectors, and $\lambda_{\text{init}} \in (0,1)$ is a constant used for the initialization of λ. DA-Router adopts a multi-head mechanism to stabilize anatomical weights.

Thus, the ROI-specific features $\boldsymbol{F}^{\text{sp}}$ are weighted by $\boldsymbol{w}_{\text{ana}}$ to derive the overall specific feature $\boldsymbol{f}^{\text{sp}} = \boldsymbol{w}_{\text{ana}}^\top \boldsymbol{F}^{\text{sp}}$. Finally, A-MoM uses a classifier $\mathrm{Cls}(\cdot)$ to predict the label $\hat{y}$ based on the concatenated overall shared and specific features:

$$\hat{y}_i = \mathrm{Cls}\left([\boldsymbol{f}^{\text{sh}}, \boldsymbol{f}^{\text{sp}}]\right), \tag{12}$$

where $[\cdot,\cdot]$ denotes row-wise (horizontal) concatenation.

Let $\boldsymbol{w}_{\text{ana}}^{\mathcal{T}}$ and $\boldsymbol{w}_{\text{ana}}^{\mathcal{S}}$ be the teacher and student anatomical routing weights. Thus, we introduce an anatomy-aware distillation loss $\mathcal{L}_{\text{anaKD}}$ that aligns their fine-grained attention distributions:

$$\mathcal{L}_{\text{anaKD}} = -\boldsymbol{w}_{\text{ana}}^{\mathcal{T}} \log\left(\boldsymbol{w}_{\text{ana}}^{\mathcal{S}}\right). \tag{13}$$

2.3. Relation-aware distillation

To learn stable between-subject relations, we propose a relation-aware distillation that performs generalizable pairwise alignment between student-related relations and the more powerful teacher–teacher relation, yielding better cross-cohort generalization, as shown in Fig. 2**(c)**. Inspired by DINO v3's Gram Anchoring (Siméoni et al., 2025), we treat any role-specific (teacher or student) feature within a batch as an *anchor*. Taking the teacher–teacher relation as an example, it captures *how each teacher anchor relates to other teacher anchors*. We then align the student–student relation $\boldsymbol{R}^{\mathcal{SS}}$ and two cross-role relations, *i.e.* student–teacher $\boldsymbol{R}^{\mathcal{ST}}$, and teacher–student $\boldsymbol{R}^{\mathcal{TS}}$, to this reference teacher–teacher relation $\boldsymbol{R}^{\mathcal{TT}}$.

Let $\boldsymbol{f}_i^{\mathcal{T}} = [\boldsymbol{f}_i^{\text{sh}(\mathcal{T})}, \boldsymbol{f}_i^{\text{sp}(\mathcal{T})}] \in \mathbb{R}^{2C_1}$ and $\boldsymbol{f}_i^{\mathcal{S}} = [\boldsymbol{f}_i^{\text{sh}(\mathcal{S})}, \boldsymbol{f}_i^{\text{sp}(\mathcal{S})}] \in \mathbb{R}^{2C_2}$ denote the overall teacher and student features, where C_1 and C_2 are their respective feature dimensions. To characterize between-subject relations, we use the normalized Gram matrix $\mathrm{Gr}(\boldsymbol{A}) = \frac{\boldsymbol{A}\boldsymbol{A}^\top}{\|\boldsymbol{A}\|\cdot\|\boldsymbol{A}\|}$ for $\boldsymbol{A} \in \mathbb{R}^{B\times C}$, where $\|\cdot\|$ denotes the l_2 norm of the column vectors, yielding a $B \times B$ pairwise similarity matrix over a minibatch of size B. Thus, the teacher–teacher and student–student relations are:

$$\boldsymbol{R}^{\mathcal{TT}} = \mathrm{Gr}\left(\mathtt{conc}(\boldsymbol{f}_1^{\mathcal{T}}, \boldsymbol{f}_2^{\mathcal{T}}, \ldots, \boldsymbol{f}_B^{\mathcal{T}})\right), \tag{14}$$

$$\boldsymbol{R}^{\mathcal{SS}} = \mathrm{Gr}\left(\mathtt{conc}(\boldsymbol{f}_1^{\mathcal{S}}, \boldsymbol{f}_2^{\mathcal{S}}, \ldots, \boldsymbol{f}_B^{\mathcal{S}})\right). \tag{15}$$

Furthermore, the teacher–student relation $\boldsymbol{R}^{\mathcal{TS}}$ and the student–teacher relation $\boldsymbol{R}^{\mathcal{ST}}$ are constructed via cross-role feature conversions to impose generalizable constraints. Specifically, to resolve dimensional mismatch and isolate ROI-shared and ROI-specific knowledge, we introduce two learnable matrices, $\boldsymbol{W}_{\text{sh}} \in \mathbb{R}^{C_1\times C_2}$ and $\boldsymbol{W}_{\text{sp}} \in \mathbb{R}^{C_1\times C_2}$, enabling bidirectional $\mathcal{S} \to \mathcal{T}$ and $\mathcal{T} \to \mathcal{S}$ conversions with tied parameters (the reverse solely uses the transpose). The teacher-to-student feature $\boldsymbol{f}_i^{\mathcal{S}\to\mathcal{T}}$ and student-to-teacher feature $\boldsymbol{f}_i^{\mathcal{T}\to\mathcal{S}}$ are then:

$$\boldsymbol{f}_i^{\mathcal{S}\to\mathcal{T}} = [\boldsymbol{f}_i^{\text{sh}(\mathcal{S})}\boldsymbol{W}_{\text{sh}}^\top, \boldsymbol{f}_i^{\text{sp}(\mathcal{S})}\boldsymbol{W}_{\text{sp}}^\top], \tag{16}$$

$$\boldsymbol{f}_i^{\mathcal{T}\to\mathcal{S}} = [\boldsymbol{f}_i^{\text{sh}(\mathcal{T})}\boldsymbol{W}_{\text{sh}}, \boldsymbol{f}_i^{\text{sp}(\mathcal{T})}\boldsymbol{W}_{\text{sp}}]. \tag{17}$$

Consequently, the corresponding relations are defined as:

$$\boldsymbol{R}^{\mathcal{ST}} = \frac{1}{B}\sum_{i=1}^{B}\mathrm{Gr}\left(\mathtt{conc}(\boldsymbol{f}_1^{\mathcal{T}},\ldots,\boldsymbol{f}_{i-1}^{\mathcal{T}},\boldsymbol{f}_i^{\mathcal{S}\to\mathcal{T}},\boldsymbol{f}_{i+1}^{\mathcal{T}},\ldots,\boldsymbol{f}_B^{\mathcal{T}})\right), \tag{18}$$

$$\boldsymbol{R}^{\mathcal{TS}} = \frac{1}{B}\sum_{i=1}^{B}\mathrm{Gr}\left(\mathtt{conc}(\boldsymbol{f}_1^{\mathcal{S}},\ldots,\boldsymbol{f}_{i-1}^{\mathcal{S}},\boldsymbol{f}_i^{\mathcal{T}\to\mathcal{S}},\boldsymbol{f}_{i+1}^{\mathcal{S}},\ldots,\boldsymbol{f}_B^{\mathcal{S}})\right). \tag{19}$$

We then use centered kernel alignment (CKA) (Alvarez, 2023) to quantify the geometric similarity between student-related relations ($\boldsymbol{R}^{\mathcal{SS}}$, $\boldsymbol{R}^{\mathcal{ST}}$) and $\boldsymbol{R}^{\mathcal{TS}}$, and the reference teacher–teacher relation $\boldsymbol{R}^{\mathcal{TT}}$, yielding the relation-aware distillation loss:

$$\begin{aligned}\mathcal{L}_{\text{relKD}} &= \mathcal{L}_{\text{relKD}}^{\mathcal{SS}} + \mathcal{L}_{\text{relKD}}^{\mathcal{ST}} + \mathcal{L}_{\text{relKD}}^{\mathcal{TS}} \\ &= \left(1 - \mathrm{CKA}(\boldsymbol{R}^{\mathcal{SS}}, \boldsymbol{R}^{\mathcal{TT}})\right) \\ &+ \left(1 - \mathrm{CKA}(\boldsymbol{R}^{\mathcal{ST}}, \boldsymbol{R}^{\mathcal{TT}})\right) \\ &+ \left(1 - \mathrm{CKA}(\boldsymbol{R}^{\mathcal{TS}}, \boldsymbol{R}^{\mathcal{TT}})\right),\end{aligned} \tag{20}$$

where $\mathrm{CKA}(\boldsymbol{G}_X, \boldsymbol{G}_Y) = \frac{\|\boldsymbol{G}_Y^\top \boldsymbol{G}_X\|^2}{\|\boldsymbol{G}_X^\top \boldsymbol{G}_X\|\cdot\|\boldsymbol{G}_Y^\top \boldsymbol{G}_Y\|}$ for Gram matrices $\boldsymbol{G}_X$ and $\boldsymbol{G}_Y$, with scores in $[0,1]$ ranging from no correlation to perfect consistency (Z. Zhou et al., 2024). By aligning pairwise relational structures via CKA, the student is guided to encode relative pathological distances robust to anatomical variability, refining its latent space toward the more faithful disease continuum geometry afforded by PET.

2.4. Overview of E^{2}AD

E^{2}AD consists of two sequential stages: pre-training a multimodal teacher and training an MRI-only student through anatomy- and relation-aware KD, as shown in Algorithm 1.

Multimodal Teacher Pre-training. We pre-train the multimodal teacher on N_1 MRI–PET pairs by minimizing the CE loss $\mathcal{L}_{\text{CE}}$ and the mapper regularization loss $\mathcal{L}_{\text{Reg}}$ in Eq. (9):

$$\mathcal{L}_{\text{Multi}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}}. \tag{21}$$

MRI-only Student Distillation. We train the MRI-only student on all $N_1 + N_2$ subjects with $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{Reg}}$ for basic discriminative capacity,

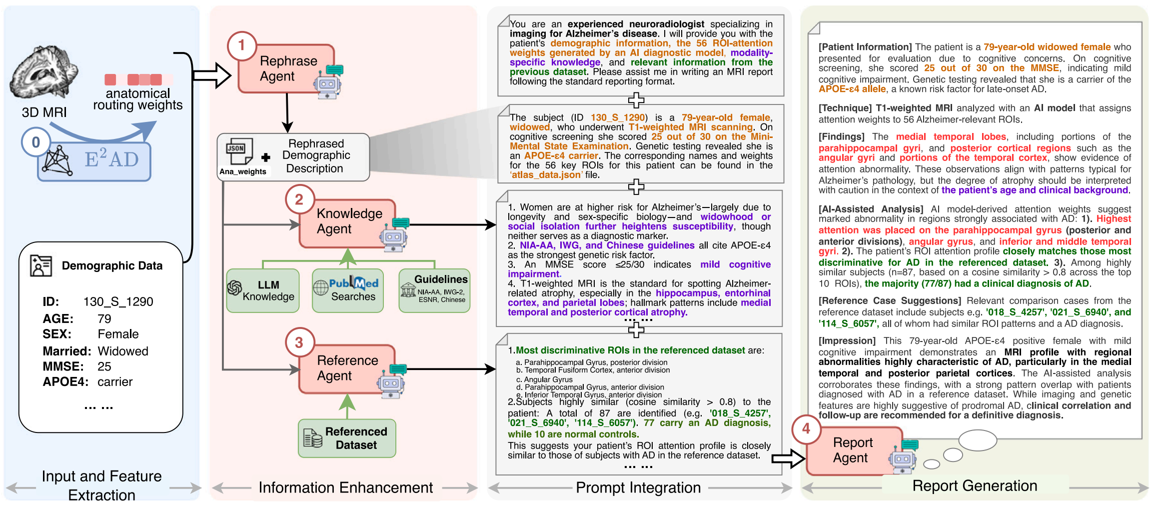

Fig. 5. Extended 3D MRI report agent workflow customized by our E^{2}AD. Our workflow employs four specialized LLM agents (rephrase, knowledge, reference, and report) that collaborate to generate the final report. The orange, red, purple, and green text respectively represent demographic-related, brain anatomy-related, established AD knowledge-related, and reference-related content.

Algorithm 1: Training procedure of E^{2}AD

Input: Multimodal dataset $\mathcal{D}_{\text{Multi}} = \{(\boldsymbol{X}_i^{\text{M}}, \boldsymbol{X}_i^{\text{P}}, y_i)\}_{i=1}^{N_1}$ and MRI-only dataset $\mathcal{D}_{\text{Single}} = \{(\boldsymbol{X}_j^{\text{M}}, y_j)\}_{j=1}^{N_2}$

Optimize: teacher parameters $\theta^{\mathcal{T}}$ and student parameters $\theta^{\mathcal{S}}$

▷ *Stage 1: Multimodal Teacher Pre-training*

for *each mini-batch* **do**

  Sample $(\boldsymbol{X}_i^{\text{M}}, \boldsymbol{X}_i^{\text{P}}, y_i)$ from $\mathcal{D}_{\text{Multi}}$;

  Calculate $\mathcal{L}_{\text{Multi}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}}$ ←Eq. (21);

  Update $\theta^{\mathcal{T}}$ via $\nabla_{\theta^{\mathcal{T}}} \mathcal{L}_{\text{Multi}}$;

end

▷ *Stage 2: MRI-only Student Distillation*

for *each mini-batch* **do**

  Sample $(\boldsymbol{X}_i^{\text{M}}, \boldsymbol{X}_i^{\text{P}}, y_i)$ from $\mathcal{D}_{\text{Multi}}$;

  Sample $(\boldsymbol{X}_j^{\text{M}}, y_j)$ from $\mathcal{D}_{\text{Single}}$;

  `// ▷ Training with all` $N_1 + N_2$ `MRI-available data`

  Calculate $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{Reg}}$;

  `// ▷ Training with` N_1 `complete MRI+ PET paired data`

  Calculate $\mathcal{L}_{\text{logitKD}}$ ← Eq. (1); ▷ *Logit-level distillation*

  Calculate $\mathcal{L}_{\text{anaKD}}$ ←Eq. (13); ▷ *Anatomy-aware distillation*

  Calculate $\mathcal{L}_{\text{relKD}}$ via CKA with $\boldsymbol{R}^{\mathcal{SS}}$, $\boldsymbol{R}^{\mathcal{ST}}$, and $\boldsymbol{R}^{\mathcal{TS}}$ to $\boldsymbol{R}^{\mathcal{TT}}$ ←Eq. (20); ▷ *Relation-aware distillation*

  Calculate $\mathcal{L}_{\text{Distill}}$ via $\mathcal{L}_{\text{logitKD}}$, $\mathcal{L}_{\text{anaKD}}$, and $\mathcal{L}_{\text{relKD}}$ ←Eq. (22);

  Calculate $\mathcal{L}_{\text{Single}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}} + \mathcal{L}_{\text{Distill}}$ ←Eq. (23);

  Update $\theta^{\mathcal{S}}$ via $\nabla_{\theta^{\mathcal{S}}} \mathcal{L}_{\text{Single}}$;

end

---

and transfer the teacher's diagnostic capability via the distillation loss $\mathcal{L}_{\text{Distill}}$ on the multimodal subset:

$$\mathcal{L}_{\text{Distill}} = \lambda_{\text{logit}} \mathcal{L}_{\text{logitKD}} + \lambda_{\text{ana}} \mathcal{L}_{\text{anaKD}} + \lambda_{\text{rel}} \mathcal{L}_{\text{relKD}}, \tag{22}$$

where $\lambda_{\text{logit}} = 1$, $\lambda_{\text{ana}} = 10$, and $\lambda_{\text{rel}} = 1$ to keep magnitudes comparable. Thus, the final student training objective is:

$$\mathcal{L}_{\text{Single}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}} + \mathcal{L}_{\text{Distill}}. \tag{23}$$

Note that the teacher and student share identical architectures except for ROI tokenization: the teacher uses two A-Toks for MRI and PET before ROI-wise feature concatenation.

At inference, we can solely use the MRI-only model to deliver superior diagnostic performance, with each prediction accompanied by a 3D MRI report generated by the multi-agent workflow described below, facilitating clinical adoption.

2.5. Clinical usage with extended multi-agent workflow

To further enhance the intuitive explainability and practical usability of E^{2}AD, we extend it, already deployable as an MRI-only model, with a compact multi-agent workflow for automatic report generation (Fig. 5). The workflow comprises four agents: (1) a *rephrase agent* that organizes patient information into a standardized input format, (2) a *knowledge agent* that integrates modality-specific and guideline-based knowledge, (3) a *reference agent* that employs retrieval-augmented generation (RAG) (Lewis et al., 2020) to surface clinically similar cases, and (4) a *report agent* that composes the final report.

The pipeline proceeds in four phases: (1) *Input and feature extraction*: given patient demographics and an MRI, E^{2}AD computes anatomical routing weights. (2) *Information enhancement*: the rephrase agent formats demographics and exports detailed routing weights (Ana_weights.json); the knowledge agent aggregates medical background from LLM summaries, PubMed searches, and established guidelines; and the reference agent compares the patient's anatomical routing weights with those from the reference (training) dataset via cosine similarity, yielding the diagnostic distribution of the retrieved cohort. (3) *Prompt integration*: outputs from all agents are merged into a single, structured prompt. (4) *Report generation*: the report agent produces a standardized diagnostic report covering patient information, technique, findings, AI-assisted analysis, and diagnostic impressions.

3. Experiment

3.1. Experimental setup

Datasets. This study contains MRI and PET from three public cohorts: Alzheimer's Disease Neuroimaging Initiative (ADNI), Australian Imaging, Biomarkers and Lifestyle Study of Ageing (AIBL), and National Alzheimer's Coordinating Center (NACC). ADNI collects multimodal data across North America to advance AD research, while AIBL is an Australian cohort, and NACC aggregates data from more than 42 U.S. AD Research Centers (ADRCs) (Beekly et al., 2004). All studies received approval from local institutional review boards, and all participants provided informed consent (Petersen et al., 2010).

Detailed demographic information for each dataset is shown in Table 1. Mild cognitive impairment (MCI) patients were classified as

Table 1
Demographic information of the subjects used in E^{2}AD. Mini-mental state examination (MMSE) and apolipoprotein E4 (APOE4) allele gene are unavailable for some subjects.

Type	Dataset	Status	Subj. Num.	Gender male (%)	Age mean (std)	MMSE mean (std)	APOE4 positive (%)
MRI	ADNI	NC	673	292 (43.39)	72.22 (6.78)	29.06 (1.15)	186 (30.95)
		sMCI	356	220 (61.8)	72.27 (7.54)	28.01 (1.65)	141 (39.61)
		pMCI	178	108 (60.67)	73.96 (7.02)	26.88 (1.7)	117 (65.73)
		AD	391	218 (55.75)	74.9 (7.87)	23.09 (2.18)	251 (66.93)
	AIBL	NC	464	198 (42.67)	72.57 (6.19)	28.68 (1.25)	126 (27.69)
		AD	77	32 (41.56)	74.01 (7.66)	20.6 (5.4)	53 (71.62)
	NACC	sMCI	155	93 (60)	72.1 (8.45)	27.85 (1.6)	59 (39.6)
		pMCI	153	86 (56.21)	75.39 (8.02)	26.78 (1.71)	83 (59.29)
PET	ADNI	NC	248	132 (53.23)	74.43 (5.81)	29.02 (1.14)	66 (26.61)
		sMCI	196	121 (61.73)	71.95 (7.63)	27.95 (1.64)	78 (39.8)
		pMCI	118	69 (58.47)	73.55 (6.91)	26.95 (1.62)	76 (64.41)
		AD	228	133 (58.33)	74.96 (7.76)	23.21 (2.13)	151 (66.52)

progressive MCI (pMCI) if they progressed to AD within 36 months, and as stable MCI (sMCI) if they did not. ADNI data are used for model development and internal testing, AIBL data for external testing of the NC vs. AD task, and, owing to the limited availability of MCI subjects, the NACC dataset for external testing of pMCI vs. sMCI. We performed stratified five-fold cross-validation on the ADNI dataset, preserving diagnostic class ratios and preventing subject overlap across folds.

Preprocessing Pipelines. All MRIs were preprocessed with a six-step pipeline (Qiu et al., 2020; Wang et al., 2024): (1) anterior and posterior commissures alignment, (2) intensity correction, (3) skull stripping, (4) outlier clipping (values $\geq 99^{th}$ percentile), (5) min–max normalization to $[-1, 1]$, and (6) center cropping to $192{\times}224{\times}192$. The first three steps were performed by FreeSurfer (Fischl, 2012). Atlas masks were obtained by FSL registration (Jenkinson et al., 2012) to the Harvard–Oxford cortical and subcortical atlases (Desikan et al., 2006), with bilateral merging and pruning to 56 ROIs (brainstem excluded). PET scans underwent five steps: (1) skull stripping with s3 algorithm (Lipková et al., 2019) in ANTs 2.1.0 (Avants et al., 2009), (2) MRI registration via FSL, and (3–5) the same outlier clipping, normalization, and center cropping as for MRI.

Implementation Details. All models were implemented in PyTorch (Paszke et al., 2019) with Kaiming initialization (He et al., 2015) and trained on NVIDIA V100 GPUs. We use CNN-based residual blocks as the voxel encoder to enhance local spatial features and reduce voxel-level registration errors (Jiang and Miao, 2024). Agents were built on DeepSeek-R1 (Guo et al., 2025). The multimodal teacher was pre-trained for 50 epochs (batch size 6, learning rate 2×10^{-4}) with a 5-epoch linear warm-up followed by exponential decay (rate 0.9). The MRI-only student was distilled for 40 epochs (batch size 8, learning rate 5×10^{-4}), with other hyperparameters unchanged. Loss hyperparameters were not tuned to enforce comparable numerical scales. Performance is evaluated by accuracy (ACC), AUC, F1-score (F1S), precision (PRE), and recall (REC), reported as percentages. Detailed ROI information and tailored LLM agent prompts are shown in Supps. A and B, respectively.

3.2. Comparison to SOTA methods

We comprehensively evaluate our E^{2}AD on AD diagnosis (NC vs. AD) and MCI progression (pMCI vs. sMCI) tasks against 13 SOTA methods, *eight published in the past three years*, grouped into five categories: (1) CNN-based: 3D ResNet (He et al., 2016), 3D SeNet (Hu et al., 2018), and I3D (Zhang et al., 2023); (2) Transformer-based: 3D Swin (Liu et al., 2022), MadFormer (Ye et al., 2024), and M3T (Jang and Hwang, 2022); (3) Anatomy-related: 3D PIPNet (De Santi et al., 2024), AAGN (Jiang and Miao, 2024), and an MRI-only variant of our E^{2}AD without cross-modal distillation, *i.e.* E^{2}AD$_{sin}$; (4) Generation-related: JointFrame (Wang et al., 2024) and ResDM (Ou et al., 2024), which employ cross-modal MRI-to-PET voxel-level synthesis to enhance diagnostic performance; and (5) Distillation-related: DCFMnet (Xiong et al., 2025) and MDT-student (Kwak et al., 2025), which directly learn more discriminative representations with the help of PET data. To ensure a fair comparison, all competing models were adjusted to ensure a comparable number of parameters.

3.2.1. Results for AD diagnosis

Table 2 summarizes the results for AD diagnosis. From these results, we highlight five key observations: (1) CNN-based methods outperform transformer-based methods, likely owing to their local inductive bias, which is particularly advantageous for large-scale 3D volumetric data. (2) Anatomy-related methods, *e.g.* 3D PIPNet, often underperform when anatomical knowledge is introduced only implicitly; by contrast, explicitly integrating brain atlas masks substantially improves performance, as evidenced by AAGN and E^{2}AD$_{sin}$. (3) Generation-related methods overemphasize voxel-level synthesis, constraining high-level discriminative capacity and yielding non-robust AUC on the external AIBL dataset; however, because AIBL is severely class-imbalanced, other metrics are less affected. (4) Distillation-related methods achieve superior performance by optimizing directly in representation space. However, despite the use of feature KD in DCFMnet and logit KD in MDT-student, such methods predominantly capture global, subject-level representations/logits and overlook anatomical priors. (5) E^{2}AD significantly outperforms all baselines on both internal ADNI and external AIBL datasets due to its hierarchical KD design, with a two-sided paired t-test confirming superiority over the second-best MDT-student, given p-values of 3.42×10^{-2} for ADNI and 1.36×10^{-2} for AIBL.

3.2.2. Results for MCI progression

Table 3 summarizes MCI progression results, showing that E^{2}AD achieves superior performance on both internal ADNI and external NACC datasets, consistent with findings from AD diagnosis. Furthermore, transformer-based and generation-related methods perform worse on MCI progression, likely due to greater pathological complexity. Among baselines, the distillation-related MDT-student emerges as the strongest through inter-modality shared information mining and logit KD. Nevertheless, E^{2}AD significantly outperforms MDT-student, as confirmed by a two-sided paired t-test, with p-values of 1.96×10^{-2} for ADNI and 1.19×10^{-2} for NACC. Please refer to Supp. C for results of multimodal teacher and Supp. D for results of specificity metrics.

3.3. Explainability with agent-generated report analysis

At the population level, the mean anatomical routing weights for each diagnostic cohort (Fig. 6(**a–b**)) reveal that the highly discriminative regions align with established AD-related neurodegeneration. Across tasks, the parahippocampal and inferior temporal gyri (Scheff et al., 2011) are consistently highlighted, reflecting their central role in AD pathophysiology. Yet task-specific contrasts are evident. AD diagnosis places greater weight on ROIs associated with advanced pathology, *e.g.* the temporal fusiform cortex (Bokde et al., 2006) and angular gyrus (Poulin et al., 2011), whereas MCI progression prioritizes ROIs

Table 2
Comparative results for AD diagnosis on internal ADNI and external AIBL datasets. I, II, III, IV, and V represent CNN-based, transformer-based, anatomy-related, generation-related, and distillation-related methods, respectively. Best results are highlighted in **bold**. (two-sided paired *t*-test: E^{2}AD vs. second-best, $p < 0.05$).

Idx.	Methods	Param.	Type	*Internal ADNI dataset*					*External AIBL dataset*				
				ACC ↑	AUC ↑	F1S ↑	PRE ↑	REC ↑	ACC ↑	AUC ↑	F1S ↑	PRE ↑	REC ↑
1	**3D ResNet**	33.1M	I	$83.2_{\pm 2.3}$	$90.1_{\pm 1.6}$	$85.2_{\pm 2.2}$	$85.5_{\pm 2.2}$	$85.0_{\pm 2.3}$	$84.6_{\pm 2.7}$	$83.5_{\pm 2.8}$	$85.8_{\pm 2.4}$	$88.0_{\pm 2.3}$	$84.5_{\pm 2.6}$
2	**3D SeNet**	34.2M		$84.7_{\pm 2.7}$	$91.0_{\pm 1.6}$	$86.0_{\pm 2.5}$	$86.2_{\pm 2.3}$	$85.8_{\pm 2.6}$	$85.0_{\pm 2.6}$	$84.2_{\pm 2.3}$	$85.8_{\pm 2.3}$	$88.4_{\pm 2.0}$	$84.4_{\pm 2.7}$
3	**I3D**	35.6M		$85.9_{\pm 2.1}$	$91.7_{\pm 1.8}$	$86.5_{\pm 1.5}$	$86.6_{\pm 1.4}$	$86.4_{\pm 1.6}$	$86.4_{\pm 2.5}$	$86.5_{\pm 1.6}$	$86.7_{\pm 1.9}$	$89.1_{\pm 1.8}$	$84.8_{\pm 2.5}$
4	**3D Swin**	30.9M	II	$82.4_{\pm 1.9}$	$87.9_{\pm 1.5}$	$83.5_{\pm 1.6}$	$83.7_{\pm 1.4}$	$83.5_{\pm 1.9}$	$81.7_{\pm 2.1}$	$80.6_{\pm 1.9}$	$82.5_{\pm 1.7}$	$85.2_{\pm 1.8}$	$80.9_{\pm 2.0}$
5	**MadFormer**	33.8M		$83.5_{\pm 2.1}$	$90.4_{\pm 1.8}$	$85.3_{\pm 2.0}$	$85.8_{\pm 1.4}$	$85.1_{\pm 2.3}$	$83.1_{\pm 2.3}$	$82.5_{\pm 1.8}$	$84.7_{\pm 2.0}$	$86.6_{\pm 1.9}$	$83.4_{\pm 2.1}$
6	**M3T**	28.4M		$82.9_{\pm 2.1}$	$89.7_{\pm 2.0}$	$85.0_{\pm 2.1}$	$85.3_{\pm 1.9}$	$84.8_{\pm 2.1}$	$82.9_{\pm 2.2}$	$82.0_{\pm 2.2}$	$84.6_{\pm 2.5}$	$87.1_{\pm 2.2}$	$83.0_{\pm 2.4}$
7	**3D PIPNet**	33.2M	III	$84.3_{\pm 2.0}$	$90.7_{\pm 1.8}$	$85.9_{\pm 1.7}$	$86.3_{\pm 1.4}$	$85.8_{\pm 1.8}$	$83.8_{\pm 2.5}$	$82.9_{\pm 2.3}$	$85.6_{\pm 2.1}$	$87.4_{\pm 1.9}$	$84.2_{\pm 2.2}$
8	**AAGN**	31.8M		$86.5_{\pm 1.2}$	$91.9_{\pm 1.5}$	$87.5_{\pm 1.3}$	$88.0_{\pm 1.3}$	$86.7_{\pm 1.5}$	$87.8_{\pm 1.3}$	$87.6_{\pm 1.6}$	$87.8_{\pm 1.4}$	$87.8_{\pm 1.4}$	$87.9_{\pm 1.4}$
9	**E^{2}AD$_{sin}$**	28.0M		$87.4_{\pm 0.9}$	$92.8_{\pm 1.2}$	$88.3_{\pm 1.0}$	$88.7_{\pm 1.0}$	$88.1_{\pm 1.1}$	$88.9_{\pm 1.1}$	$88.2_{\pm 1.3}$	$88.5_{\pm 1.2}$	$88.3_{\pm 1.2}$	$88.9_{\pm 1.2}$
10	**JointFrame**	30.4M	IV	$86.6_{\pm 1.8}$	$92.1_{\pm 2.2}$	$87.7_{\pm 2.1}$	$87.9_{\pm 2.1}$	$87.6_{\pm 1.9}$	$87.4_{\pm 2.0}$	$87.1_{\pm 2.0}$	$87.4_{\pm 2.1}$	$88.7_{\pm 2.1}$	$86.3_{\pm 2.3}$
11	**ResDM**	33.2M		$87.5_{\pm 2.1}$	$92.8_{\pm 1.9}$	$88.4_{\pm 2.0}$	$89.0_{\pm 1.9}$	$87.5_{\pm 2.1}$	$87.8_{\pm 2.4}$	$87.4_{\pm 2.4}$	$87.7_{\pm 2.1}$	$88.7_{\pm 2.5}$	$86.9_{\pm 1.7}$
12	**DCFMnet**	44.5M	V	$86.8_{\pm 1.9}$	$92.2_{\pm 1.7}$	$87.9_{\pm 1.8}$	$89.3_{\pm 1.7}$	$86.8_{\pm 1.9}$	$87.5_{\pm 1.9}$	$87.1_{\pm 1.8}$	$87.4_{\pm 2.0}$	$87.9_{\pm 1.9}$	$87.1_{\pm 2.2}$
13	**MDT-student**	33.1M		$89.0_{\pm 1.2}$	$93.4_{\pm 1.5}$	$89.2_{\pm 1.1}$	$89.4_{\pm 1.1}$	$89.0_{\pm 1.3}$	$89.4_{\pm 2.0}$	$89.4_{\pm 1.5}$	$89.0_{\pm 1.7}$	$89.4_{\pm 1.2}$	$88.6_{\pm 2.0}$
	E^{2}AD (ours)	28.0M		$\mathbf{90.3}_{\pm 1.1}$	$\mathbf{94.2}_{\pm 1.1}$	$\mathbf{90.0}_{\pm 1.3}$	$\mathbf{90.2}_{\pm 1.4}$	$\mathbf{89.8}_{\pm 1.3}$	$\mathbf{91.0}_{\pm 0.8}$	$\mathbf{90.6}_{\pm 0.7}$	$\mathbf{90.5}_{\pm 0.9}$	$\mathbf{90.7}_{\pm 0.5}$	$\mathbf{90.1}_{\pm 0.5}$

Table 3
Comparative results for MCI progression prediction on internal ADNI and external NACC datasets. I, II, III, IV, and V represent CNN-based, transformer-based, anatomy-related, generation-related, and distillation-related methods, respectively. Best results are highlighted in **bold**. (two-sided paired *t*-test: E^{2}AD vs. second-best, $p < 0.05$).

Idx.	Methods	Param.	Type	*Internal ADNI dataset*					*External NACC dataset*				
				ACC ↑	AUC ↑	F1S ↑	PRE ↑	REC ↑	ACC ↑	AUC ↑	F1S ↑	PRE ↑	REC ↑
1	**3D ResNet**	33.1M	I	$65.6_{\pm 3.1}$	$66.0_{\pm 2.5}$	$65.3_{\pm 2.1}$	$65.6_{\pm 2.7}$	$65.3_{\pm 2.4}$	$55.2_{\pm 3.2}$	$60.1_{\pm 2.9}$	$56.5_{\pm 2.2}$	$57.3_{\pm 2.4}$	$55.2_{\pm 2.4}$
2	**3D SeNet**	34.2M		$65.9_{\pm 2.9}$	$66.2_{\pm 1.8}$	$65.7_{\pm 2.5}$	$65.9_{\pm 2.1}$	$65.5_{\pm 2.9}$	$56.6_{\pm 2.8}$	$60.6_{\pm 2.6}$	$57.4_{\pm 2.4}$	$57.2_{\pm 2.5}$	$57.6_{\pm 2.6}$
3	**I3D**	35.6M		$66.6_{\pm 2.5}$	$67.5_{\pm 2.6}$	$66.2_{\pm 2.2}$	$66.5_{\pm 2.2}$	$65.9_{\pm 2.3}$	$58.3_{\pm 2.8}$	$61.7_{\pm 2.7}$	$60.4_{\pm 2.3}$	$60.7_{\pm 2.2}$	$60.5_{\pm 2.5}$
4	**3D Swin**	30.9M	II	$61.2_{\pm 2.4}$	$62.6_{\pm 2.4}$	$61.8_{\pm 2.3}$	$62.4_{\pm 2.8}$	$61.3_{\pm 3.0}$	$52.6_{\pm 2.6}$	$56.6_{\pm 2.6}$	$55.3_{\pm 2.6}$	$55.7_{\pm 3.0}$	$54.9_{\pm 2.5}$
5	**MadFormer**	33.8M		$64.9_{\pm 2.8}$	$65.8_{\pm 1.9}$	$64.8_{\pm 2.2}$	$65.5_{\pm 1.9}$	$64.7_{\pm 2.0}$	$56.0_{\pm 2.4}$	$59.3_{\pm 2.1}$	$58.2_{\pm 2.3}$	$58.7_{\pm 2.2}$	$58.0_{\pm 2.5}$
6	**M3T**	28.4M		$62.5_{\pm 3.0}$	$63.3_{\pm 2.8}$	$62.5_{\pm 1.9}$	$62.5_{\pm 2.3}$	$62.4_{\pm 2.4}$	$54.8_{\pm 3.3}$	$57.8_{\pm 2.1}$	$55.5_{\pm 2.4}$	$57.0_{\pm 2.5}$	$54.3_{\pm 2.4}$
7	**3D PIPNet**	33.1M	III	$66.0_{\pm 2.9}$	$66.3_{\pm 2.0}$	$65.7_{\pm 2.1}$	$66.1_{\pm 2.2}$	$65.4_{\pm 2.1}$	$54.6_{\pm 2.6}$	$59.8_{\pm 2.4}$	$55.9_{\pm 2.1}$	$57.4_{\pm 2.2}$	$54.2_{\pm 2.3}$
8	**AAGN**	31.8M		$69.8_{\pm 2.2}$	$70.7_{\pm 1.8}$	$69.1_{\pm 2.1}$	$69.4_{\pm 2.4}$	$68.9_{\pm 1.9}$	$62.4_{\pm 2.3}$	$65.4_{\pm 1.9}$	$64.6_{\pm 2.1}$	$65.4_{\pm 2.1}$	$63.9_{\pm 2.0}$
9	**E^{2}AD$_{sin}$**	28.0M		$71.4_{\pm 2.0}$	$71.6_{\pm 1.6}$	$71.1_{\pm 1.8}$	$71.3_{\pm 1.6}$	$70.9_{\pm 1.9}$	$64.2_{\pm 2.2}$	$66.7_{\pm 1.9}$	$65.1_{\pm 1.8}$	$65.6_{\pm 1.8}$	$65.2_{\pm 2.1}$
10	**JointFrame**	30.4M	IV	$69.6_{\pm 2.3}$	$70.4_{\pm 2.1}$	$69.0_{\pm 2.5}$	$69.2_{\pm 2.4}$	$69.0_{\pm 2.7}$	$62.0_{\pm 2.5}$	$65.3_{\pm 2.3}$	$64.5_{\pm 2.2}$	$65.7_{\pm 2.2}$	$63.5_{\pm 2.4}$
11	**ResDM**	33.2M		$70.4_{\pm 2.4}$	$71.2_{\pm 2.0}$	$69.9_{\pm 2.2}$	$70.3_{\pm 2.0}$	$69.7_{\pm 2.2}$	$62.6_{\pm 2.3}$	$65.5_{\pm 2.0}$	$65.0_{\pm 2.0}$	$65.3_{\pm 1.8}$	$64.8_{\pm 2.1}$
12	**DCFMnet**	44.5M	V	$71.5_{\pm 2.4}$	$71.7_{\pm 2.1}$	$71.5_{\pm 2.0}$	$71.8_{\pm 2.2}$	$71.3_{\pm 2.2}$	$64.4_{\pm 2.0}$	$66.8_{\pm 1.9}$	$65.3_{\pm 2.0}$	$65.7_{\pm 2.0}$	$64.8_{\pm 2.1}$
13	**MDT-student**	33.1M		$72.6_{\pm 2.8}$	$72.8_{\pm 2.7}$	$72.4_{\pm 2.6}$	$72.7_{\pm 2.3}$	$72.2_{\pm 2.5}$	$66.3_{\pm 2.1}$	$67.7_{\pm 2.0}$	$66.8_{\pm 1.9}$	$67.1_{\pm 2.1}$	$66.5_{\pm 1.9}$
	E^{2}AD (ours)	28.0M		$\mathbf{73.9}_{\pm 2.1}$	$\mathbf{74.1}_{\pm 1.9}$	$\mathbf{73.8}_{\pm 1.8}$	$\mathbf{74.0}_{\pm 1.6}$	$\mathbf{73.6}_{\pm 2.1}$	$\mathbf{68.0}_{\pm 2.0}$	$\mathbf{69.3}_{\pm 1.9}$	$\mathbf{68.6}_{\pm 1.8}$	$\mathbf{68.8}_{\pm 2.0}$	$\mathbf{68.4}_{\pm 1.7}$

implicated in earlier changes and subtle cognitive decline, *e.g.* the cingulate gyrus (Jones et al., 2006) and superior temporal gyrus (Liu et al., 2013). Furthermore, retraining our E^{2}AD across five random seeds yields highly consistent top-5 routing weights, verifying the population-level reliability of our DA-Router. At the individual level, Fig. 6(**c**) presents a random AD case, where E^{2}AD assigns weights in proportion to observed anatomical alterations: most ROIs cluster near a uniform baseline of 0.0178 (1 divided by the number of brain ROIs), ROIs with greater atrophy receive higher weights, and frontal lobe ROIs, which are typically less affected in AD, receive lower weights. Notably, rather than relying solely on the fixed ROI boundaries of a static atlas, our DA-Router dynamically re-weights each ROI, overcoming atlas rigidity to capture these stage-specific biomarkers.

An example MRI report is shown in the right panel of Fig. 5, with corresponding anatomical routing weights shown in Fig. 6(**c**). Our agent workflow demonstrates advanced clinical reasoning by integrating multimodal data and leveraging diverse knowledge sources. (1) The findings section converts anatomical routing weights into effective textual descriptions validated by radiologists, while incorporating contextual clinical judgment, including age-related confounders and psychosocial factors *e.g.* widowhood. (2) The AI-assisted analysis section provides a comprehensive evaluation based on established AD knowledge and retrieved reference information, *e.g.* 77 out of 87 similar subjects had an AD diagnosis, while offering relevant case IDs for reference. (3) The final impression provides an overall assessment with graduated confidence levels.

We conducted a subjective evaluation to compare our generated reports against the SOTA AutoRG-Brain (Lei et al., 2025). A board-certified neuroradiologist (12 years' experience) evaluated 20 randomly sampled cases under a strict double-blinded manner. Specifically, the neuroradiologist was provided only with demographic data and MRI scans, evaluating anonymized reports in a randomized order.

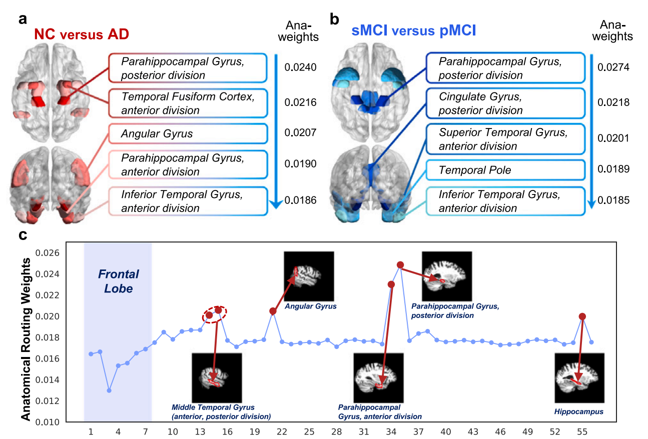

Fig. 6. Explainable results of E^{2}AD. (a–b) Top-5 population-level anatomical routing weights in the AD diagnosis and MCI progression tasks, respectively. (c) Individual-level result for a random AD patient.

The assessment utilized a 5-point Likert scale across four dimensions. These included consistency with imaging findings, diagnostic reasoning, structure and readability, and terminology standardization. As Fig. 7 demonstrates, E^{2}AD more faithfully captures fine-scale anatomy and clinically salient abnormalities, resulting in evidence-based rather than speculative descriptions. Conversely, AutoRG-Brain often misidentifies imaging modalities and occasionally hallucinates nonexistent findings (*e.g.* tumors). By conditioning on anatomical routing weights and leveraging a multi-agent referencing architecture, E^{2}AD largely reduces these hallucinations to produce highly structured, clinician-friendly

Table 4
Ablation results of different distillation components for AD diagnosis. Best results are highlighted in **bold**.

Idx.	*LogitKD*	*AnaKD*		*RelKD*			ACC↑	AUC ↑	F1S ↑	PRE ↑	REC ↑
	$\mathcal{L}_{\text{logitKD}}$	A-MoM	$\mathcal{L}_{\text{anaKD}}$	$\mathcal{L}_{\text{relKD}}^{\mathcal{SS}}$	$\mathcal{L}_{\text{relKD}}^{\mathcal{ST}}$	$\mathcal{L}_{\text{relKD}}^{\mathcal{TS}}$					
1	–	–	–	–	–	–	83.2±2.3	90.1±1.6	85.2±2.2	85.5±2.2	85.0±2.3
2	✓	–	–	–	–	–	84.5±2.1	91.5±1.4	86.8±1.9	87.1±1.5	86.5±1.8
3	✓	✓	–	–	–	–	87.8±0.8	93.0±1.0	88.6±1.0	88.7±1.1	88.5±1.0
4	✓	✓	✓	–	–	–	88.6±1.1	93.3±1.3	88.9±1.1	89.2±1.1	88.7±1.2
5	✓	✓	✓	✓	–	–	89.1±1.0	93.6±1.1	89.3±1.2	89.4±1.3	89.2±1.1
6	✓	✓	✓	–	✓	–	88.9±1.2	93.4±1.0	89.2±1.3	89.2±1.3	88.9±1.3
7	✓	✓	✓	–	–	✓	89.4±1.3	93.8±1.2	89.4±1.0	89.7±1.2	89.3±1.2
8	✓	✓	✓	✓	✓	✓	**90.3**±1.1	**94.2**±1.1	**90.0**±1.3	**90.2**±1.4	**89.8**±1.3

Table 5
Ablation study on ROI granularity for AD diagnosis across internal and external datasets. Best results are highlighted in **bold**.

Variant		**Parcellation Type**	**ROI Number**	*Internal ADNI dataset*					*External AIBL dataset*				
				ACC ↑	**AUC ↑**	**F1S ↑**	**PRE ↑**	**REC ↑**	**ACC ↑**	**AUC ↑**	**F1S ↑**	**PRE ↑**	**REC ↑**
V1	Coarse	Lobe	5	87.5±1.6	91.8±1.5	87.2±1.5	87.4±1.6	87.0±1.7	88.5±1.4	88.2±1.5	88.1±1.4	88.3±1.3	87.8±1.6
V2	Fine	AAL	116	88.9±1.4	92.5±1.3	88.6±1.4	88.8±1.3	88.5±1.5	89.1±1.1	88.6±1.2	88.5±1.2	88.6±1.1	88.3±1.3
V3	**Ours**	**Harvard-Oxford**	**56**	**90.3**±1.1	**94.2**±1.1	**90.0**±1.3	**90.2**±1.4	**89.8**±1.3	**91.0**±0.8	**90.6**±0.7	**90.5**±0.9	**90.7**±0.5	**90.1**±0.5

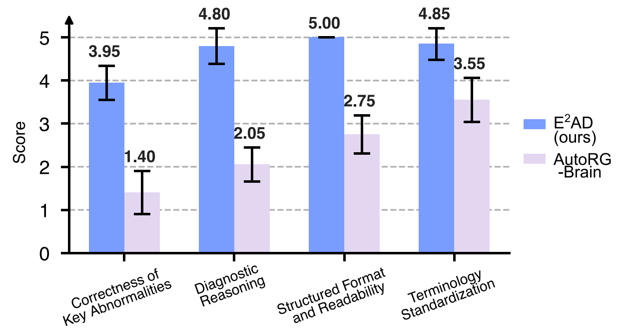

Fig. 7. Subjective evaluation of reports generated by the proposed method and AutoRG-Brain.

reports. Nonetheless, we acknowledge that the resulting recommendations remain relatively generic and are insufficiently personalized to individual patients.

3.4. Ablation study

Unless otherwise specified, all ablation studies are conducted on the internal ADNI dataset for AD diagnosis.

3.4.1. Ablation on different distillation components

Table 4 summarizes the ablation results for different distillation components, while Fig. 1(**b**) visualizes the performance gain of E^{2}AD over the baseline (3D ResNet) on AD diagnosis and MCI progression tasks. Conventional *logit-level KD* (Idx. 2) yields modest improvements due to weak supervisory signals. The *anatomy-aware KD* integrates anatomical priors through two complementary components: (i) A-MoM, which replaces the last two residual blocks of the 3D ResNet backbone to form the A-MoM architecture (Idx. 3); and (ii) a subsequent distillation stage that transfers anatomical knowledge from a multimodal teacher (Idx. 4). These two components jointly reinforce anatomical representation learning, resulting in consistent performance gains. Building on this, we examine the *relation-aware KD* by evaluating the individual contributions of $\mathcal{L}_{\text{relKD}}^{\mathcal{SS}}$, $\mathcal{L}_{\text{relKD}}^{\mathcal{ST}}$, and $\mathcal{L}_{\text{relKD}}^{\mathcal{TS}}$ (Idxs. 5-7). Among the relational terms, $\mathcal{L}_{\text{relKD}}^{\mathcal{TS}}$ proves most effective via direct cross-role backpropagation. Combining all $\mathcal{L}_{\text{relKD}}$ terms (Idx. 8) consistently yields the best diagnostic performance. These results confirm that *AnaKD* and *RelKD* provide complementary rather than overlapping supervision: AnaKD alone yields measurable gains, and further incorporating RelKD brings additional improvements. More details on the training dynamics of *AnaKD* and *RelKD* are presented in Supp. G.

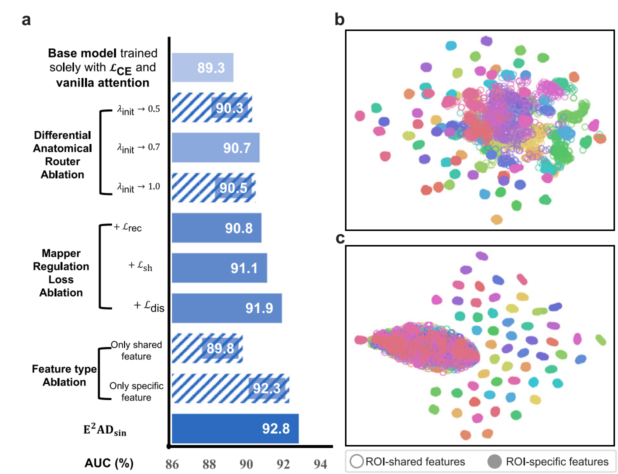

Fig. 8. Ablation study of E^{2}AD$_{\text{sin}}$. (a) Modernization results for E^{2}AD$_{\text{sin}}$; solid colors denote the selected configurations, while slashed lines denote those not adopted. (b–c) *t*-SNE visualizations of A-MoM features trained (b) without and (c) with the mapper regulation loss; colors indicate different ROIs.

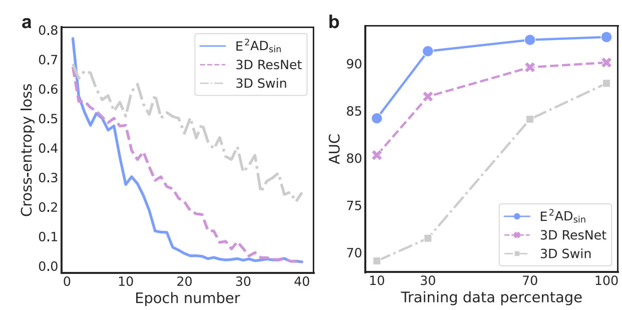

Fig. 9. Comparison of E^{2}AD$_{\text{sin}}$ with competing methods: (a) training convergence and (b) AUC curves across varying training data sizes.

3.4.2. Ablation on A-MoM

Impact of ROI Granularity. To investigate the effect of ROI parcellation granularity on diagnostic performance and cross-domain generalizability, we evaluate three variants: (V1) coarse 5-lobe parcellation, (V2) fine-grained 116-region AAL atlas (Tzourio-Mazoyer et al., 2002), and (V3) our default 56-region Harvard-Oxford atlas (Desikan et al., 2006). All variants are evaluated on both the internal ADNI and external AIBL datasets for the AD diagnostic task. Based on the results in Table 5, we have the following observations. (**i**) V1 fails to capture the localized pathological specificity of AD (*e.g.* subtle hippocampal atrophy),

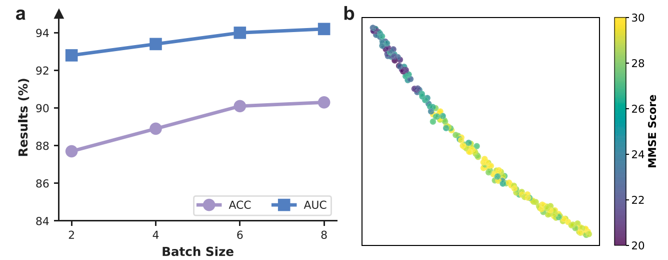

Fig. 10. Batch size sensitivity and latent visualization. (a) Diagnostic performance (ACC/AUC) across varying batch sizes. **(b)** *t*-SNE visualization; colors indicate MMSE scores.

yielding inferior performance and confirming that a minimum level of spatial granularity is essential for extracting diagnostically relevant features. **(ii)** The overly fine-grained atlas (V2) does not further improve but rather degrades performance, likely because excessive fragmentation amplifies inter-subject anatomical variance and registration errors, introducing spatial noise that hinders stable representation learning. **(iii)** V3 achieves the best results on both datasets and the smallest cross-dataset AUC drop (94.2→90.6) compared to V2's more pronounced gap (92.5→88.6), confirming that an intermediate granularity balances local spatial sensitivity and cross-domain resilience, and justifying its adoption as the default parcellation in E^{2}AD. Note that AIBL exhibits class imbalance toward NC subjects, making AUC the reliable metric.

Three complementary design choices collectively confer robustness to E^{2}AD against both atlas granularity variation and registration inaccuracies. (i) *Local spatial tolerance* is ensured by the CNN-based voxel encoder, which aggregates regional context via its receptive field, rendering features robust to minor boundary misalignments. (ii) *Topological regularization* is achieved by treating the atlas as a structural prior, steering the network toward clinically meaningful patterns and suppressing cohort-specific overfitting. (iii) *Global failsafe* is provided by the shared mapper, which operates independently of ROI boundaries to capture holistic atrophy patterns, guaranteeing a reliable performance floor under severe registration failures. Together, these mechanisms confer consistent gains across atlases of varying granularity. More boundary perturbation analysis is provided in Supp. E.

Effectiveness of Anatomical Modules. Fig. 8**(a)** presents modernization results for E^{2}AD$_{\text{sin}}$ *without* cross-modal distillation. We begin with a base model using anatomical mappers trained solely with $\mathcal{L}_{\text{CE}}$ and vanilla attention (Vaswani et al., 2017) as the anatomical routing module. Our DA-Router consistently enhances diagnostic performance by yielding more stable routing weights across λ_{init} values, with $\lambda_{\text{init}} = 0.7$ achieving the best. Incorporating the mapper regulation loss further boosts performance, with the disentanglement loss providing a notable improvement. To illustrate its effect, we visualize features using *t*-SNE without and with mapper regulation loss in Figs. 8**(b–c)**. The results show that this loss compacts ROI-shared features while promoting clearer separation of ROI-specific features. These results also highlight the use of ROI-shared features for two purposes: (i) capturing global, ROI-agnostic disease directions and (ii) enhancing ROI-specific discriminability to support anatomy-aware fusion. Feature-type ablation further indicates that relying solely on shared or specific features degrades performance, confirming their complementary roles.

Data and Training Efficiency. Compared to 3D ResNet and 3D Swin, E^{2}AD$_{\text{sin}}$ exhibits superior convergence and data efficiency (Fig. 9). In Fig. 9**(a)**, E^{2}AD$_{\text{sin}}$ converges faster than all baselines within 40 training epochs, while 3D Swin lags due to limited inductive bias. In Fig. 9**(b)**, E^{2}AD$_{\text{sin}}$ achieves competitive performance with only 10% training data, outperforming 3D ResNet, while 3D Swin degrades sharply.

3.4.3. *Further analysis of relation-aware distillation*

Impact of Batch Size. We evaluate the sensitivity of *RelKD* to batch size, which governs the local context for relational modeling. In Fig. 10**(a)**, classification performance (ACC/AUC) improves consistently as batch size increases from 2 to 8. Given the computational constraints of high-dimensional 3D MRI, a batch size of 8 approaches our hardware limit; nonetheless, performance plateaus beyond 6, confirming that this scale suffices for stable relational KD. Further quantitative analysis regarding batch composition is presented in Supp. F.

Global Disease Manifold Consistency. To verify that batch-level distillation aggregates into a coherent disease progression manifold, we visualize learned features via *t*-SNE colored by MMSE scores in Fig. 10**(b)**. The resulting visualization exhibits a continuous biological spectrum aligned with clinical cognitive trajectories, confirming that mini-batch CKA alignment grounds representations in disease biology rather than stochastic batch composition.

4. Discussion

Discussion on Related Works. MRI-based AD detection has followed three main directions. **(i)** *Multimodal fusion methods* (Wu et al., 2024; Qiu et al., 2024; H. Zhou et al., 2024; Zhang et al., 2024) jointly model MRI and PET (or other modalities) and achieve strong performance, but they assume all modalities are available at test time and degrade sharply when PET is missing, common in routine care. **(ii)** *Modality-completion methods* (Pan et al., 2020; Ou et al., 2024; Wang et al., 2024; Xu et al., 2025) synthesize PET from MRI, but typically trained to maximize voxel-level fidelity rather than diagnostic utility, making them brittle under shifts in PET reconstruction. **(iii)** *Cross-modal distillation frameworks* (Xiong et al., 2025; Liu et al., 2025; Kwak et al., 2025) transfer knowledge from a multimodal teacher to an MRI-only student, yet most act on logits or features, collapsing rich MRI–PET complementarity and ignoring neuroanatomy and population structure.

E^{2}AD advances this distillation line by unifying cross-modal knowledge transfer with explainable clinical reporting. An atlas-guided A-MoM makes anatomy explicit, while CKA-based relation-aware alignment transfers cohort-level geometry from teacher to student, converting multimodal supervision into an MRI-only model that is anatomically and relationally grounded. By distilling within-subject and between-subject priors into the model (clinical→AI) and turning its outputs into actionable MRI reports (AI→clinical), E^{2}AD establishes a bidirectional loop where clinical knowledge improves AI performance and AI explanations enhance trust in AD assessment.

Relation to Regularization-based KD Methods. Regularization-based KD methods have demonstrated consistent gains in model robustness and generalization in medical image analysis, particularly in data-scarce and modality-imbalanced settings. Notable examples include confidence-regularized KD for brain age prediction (Yang et al., 2023), patch-level KD with explicit regularization for missing-modality robustness in medical image segmentation (R. Wang et al., 2024), and dynamic modality reweighting as an implicit regularizer within multimodal KD for AD recognition (Dong et al., 2025). E^{2}AD aligns naturally with this line of work, as its mapper regulation loss and CKA-based Gram matrix alignment serve as structural and relational regularizers embedded within the KD process.

Specialization of Shared and Specific Mappers. The dual-mapper design of A-MoM is motivated by a fundamental duality in AD pathology: diffuse cortical atrophy coexists with focal, region-specific neurodegeneration (Holland et al., 2009; Poulakis et al., 2018). Accordingly, the *shared mapper* captures ROI-agnostic morphometric trends, with the shared regulation loss $\mathcal{L}_{\text{sh}}$ anchoring them to the anatomical mean. The *specific mappers*, by contrast, are driven by the disentanglement loss $\mathcal{L}_{\text{dis}}$ to isolate local heterogeneity while suppressing redundant global information. This decoupling is empirically confirmed by the *t*-SNE analysis in Fig. 8**(c)**: shared features collapse into a cohesive cluster, whereas specific features organize into well-separated, ROI-discriminative clusters. By jointly encoding global and local facets of disease progression, this design enables a more complete characterization of the AD trajectory.

Reliability of Automated Reporting. Generated reports are designed as a second-opinion aid that supports rather than replaces clinical judgment, with the practical goal of streamlining reporting workflows and reducing the burden of repetitive drafting in routine care. To mitigate LLM hallucinations and ensure reliability, we adopt three complementary strategies: **(i)** structured prompts that constrain generative freedom; **(ii)** conditioning on *anatomical routing weights* rather than raw images, tightly coupling generated text to extracted imaging features; and **(iii)** specialized multi-agent roles, wherein a knowledge agent grounds findings in clinical guidelines and a reference agent draws on RAG to incorporate evidence from relevant cases. Together, this knowledge-grounded architecture substantially reduces hallucinations compared to SOTA end-to-end generators like AutoRG-Brain (Lei et al., 2025).

Discussion on Advantages. E^2AD is a practically deployable MRI-only method that leverages multimodal supervision to provide anatomically grounded, robust, and clinically interpretable AD assessment, with five key advantages. **(i)** *Robust MRI-only performance.* Since existing methods are typically evaluated on heterogeneous cohorts under inconsistent protocols, we re-implemented all baselines under a unified protocol across ADNI, AIBL, and NACC. E^2AD consistently outperforms recent CNN-, transformer-, anatomy-aware, generative, and distillation-based methods while converging faster and using data more efficiently. **(ii)** *Anatomy-aware disease modeling.* An atlas-guided A-MoM converts encoder features into ROI-wise tokens with shared and specific components, enabling the student to learn structured region-level disease representations instead of merely matching global logits. **(iii)** *Cohort-aware latent stability.* Relation-aware supervision aligns student-related Gram matrices with a teacher–teacher reference via CKA, transferring both case-wise signals and population-level geometry, yielding more stable latent spaces across cohorts and scanners. **(iv)** *Clinically explainable reporting.* Anatomical routing weights expose disease-relevant ROIs at group and subject levels, and a multi-agent LLM uses them to generate structured MRI reports, positioning E^2AD as a decision-support tool rather than a black-box classifier. **(v)** *Practical real-world deployment.* Because inference relies only on MRI and the model is data-efficient and robust to cohort shifts, E^2AD is well suited for institutions without routine PET or large labeled cohorts and integrates into reporting workflows and multicenter studies.

Discussion on Limitations. Several limitations warrant discussion. **(i)** E^2AD relies on a predefined atlas and accurate inter-subject registration to impose neuroanatomical structure, which may reduce robustness under atypical anatomy (*e.g.* large lesions or prior surgery) and across heterogeneous acquisition protocols. Future work will explore atlas-light or atlas-free variants that learn anatomical priors in a self-supervised manner and explicitly evaluate robustness to registration noise and scanner shifts. Furthermore, although our DA-Router adaptively weights regional importance, relying on a single fixed atlas may fail to fully capture stage-dependent atrophy. Thus, developing dynamic, stage-adaptive parcellation is a valuable direction for future work. **(ii)** The multimodal teacher is trained on paired MRI-FDG-PET data from ADNI and evaluated mainly on AD and MCI populations. The extent to which the learned anatomical and relational priors transfer across alternative tracers (*e.g.* amyloid or tau PET), other neurodegenerative conditions, and diverse clinical cohorts remains to be established, necessitating larger, multi-center, and prospective studies. **(iii)** The multi-agent LLM workflow, while more grounded than single VLMs and less prone to hallucination, operates retrospectively and lacks formal uncertainty estimates. Safety-aware integration with human oversight and well-calibrated confidence will be a prerequisite for responsible clinical deployment. **(iv)** The distillation stage currently requires paired MRI–PET data, which may not be available in all institutions. Exploring weaker supervision, *e.g.* semi-supervised or federated distillation, could further lower barriers to adoption.

5. Conclusion

This paper presents E^2AD, an effective and explainable AD detection framework based on anatomy- and relation-aware cross-modal distillation to better exploit multimodal supervision. E^2AD is trained on paired MRI–PET data but deployed via widely accessible MRI only. From a *within-subject* perspective, it transfers anatomical dependencies from teacher to student via an anatomical Mixture-of-Mappers, yielding spatially meaningful and clinically traceable cues. From a *between-subject* perspective, it leverages generalizable pairwise alignment of student-related relations to the teacher–teacher relation, yielding better cross-cohort generalization. In addition, a multi-agent LLM workflow translates E^2AD's anatomical attention patterns into clinician-readable MRI reports. By integrating within-subject anatomical cues, between-subject structural relations, and agent-driven reporting, E^2AD delivers more effective and explainable MRI-only detection.

CRediT authorship contribution statement

Chenhui Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Sirong Piao:** Writing – review & editing, Data curation. **Zhihao Chen:** Writing – review & editing, Validation, Formal analysis. **Tao Chen:** Writing – review & editing, Visualization, Validation, Data curation. **Zhaoyang Li:** Writing – review & editing, Validation, Data curation. **Tongrui Zhang:** Writing – review & editing, Visualization, Validation. **Yuxin Li:** Writing – review & editing, Supervision. **Xing-Ming Zhao:** Writing – review & editing, Supervision, Resources. **Hong-ming Shan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.media.2026.104099.

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